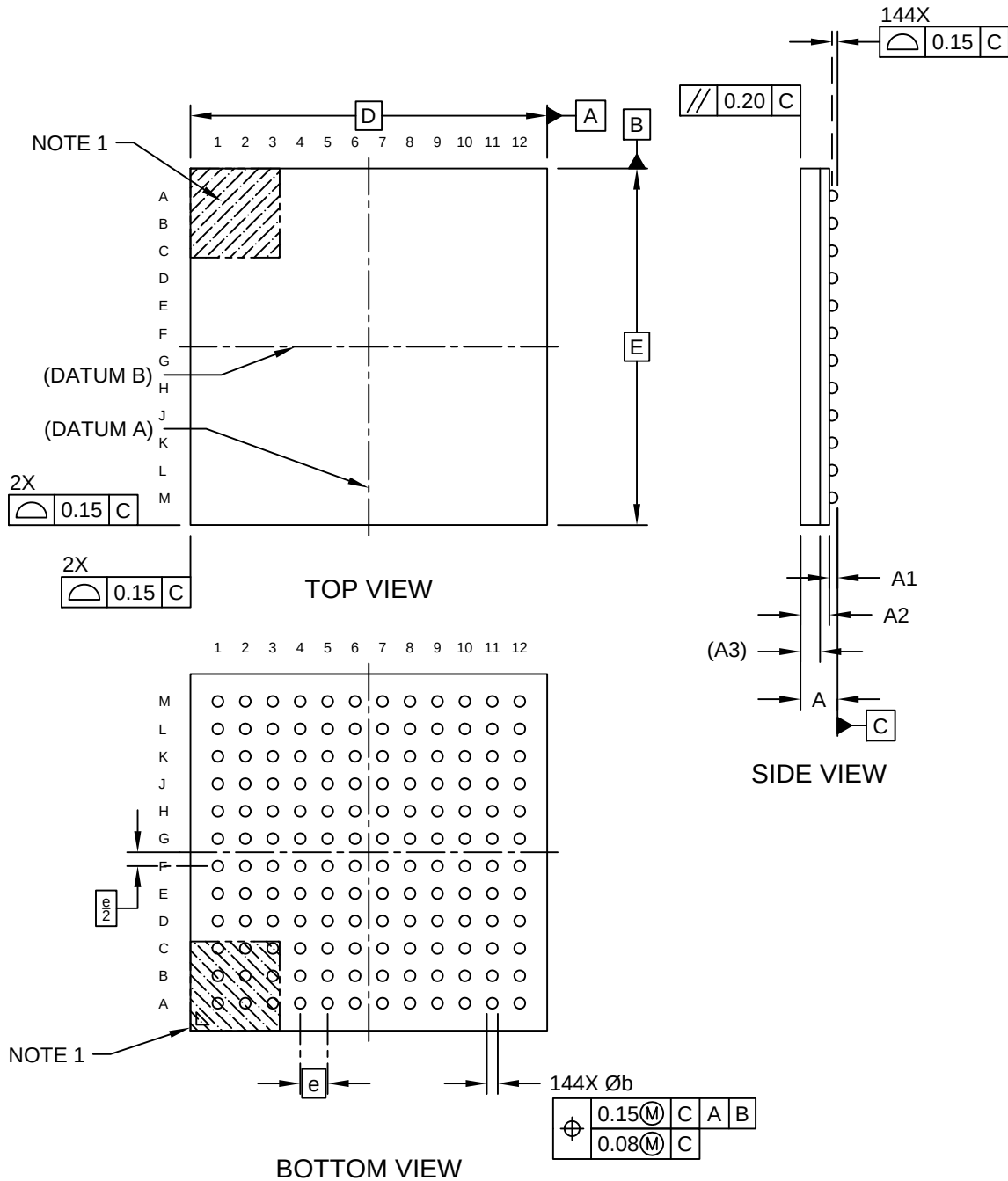


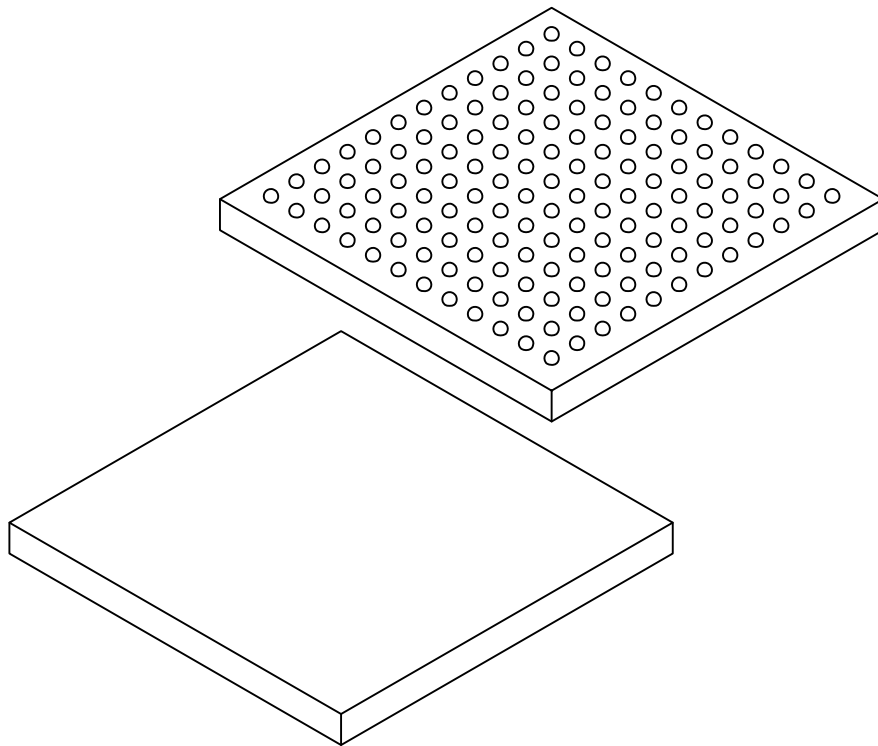
144-Ball Low Profile Ball Grid Array (A4B) - 13x13x1.35 mm Body [LBGA] Atmel Legacy Global Package Code CXK

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**144-Ball Low Profile Ball Grid Array (A4B) - 13x13x1.35 mm Body [LBGA]
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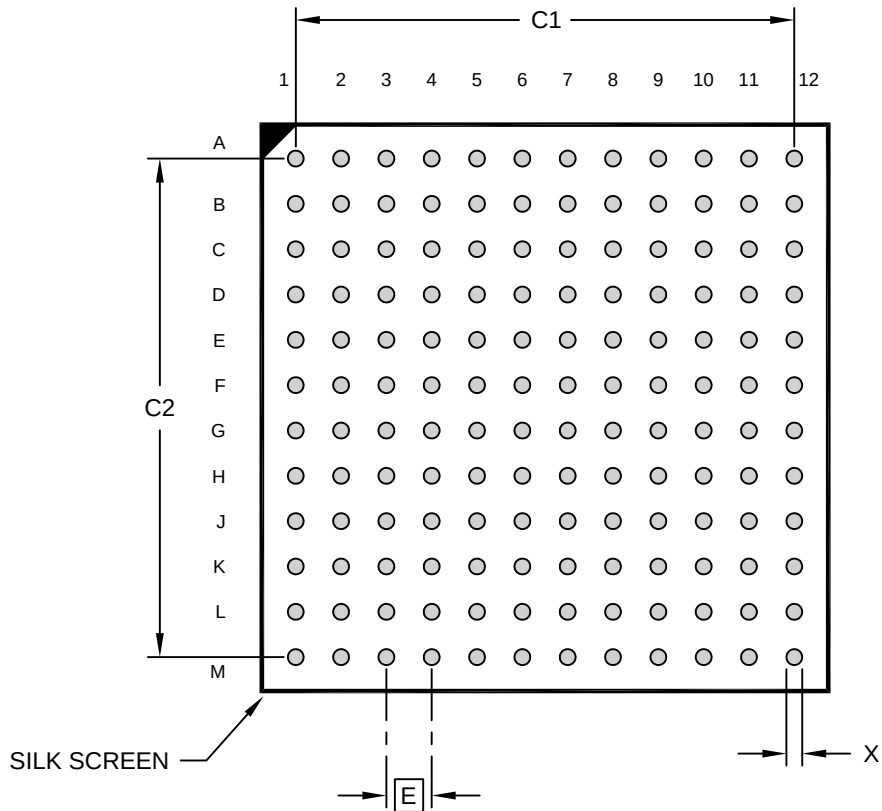
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	144		
Pitch	e	1.00 BSC		
Overall Height	A	-	-	1.35
Standoff	A1	0.25	0.30	0.35
Terminal Thickness	A2	0.92	0.96	1.00
Mold Cap Thickness	A3	0.70 REF		
Overall Length	D	13.00 BSC		
Overall Width	E	13.00 BSC		
Terminal Diameter	b	0.35	0.40	0.45

Notes:

1. Terminal A1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	1.00 BSC		
Overall Pitch	C1		11.00	
Overall Pitch	C2		11.00	
Contact Pad Diameter (X144)	X			0.35

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23112 Rev A